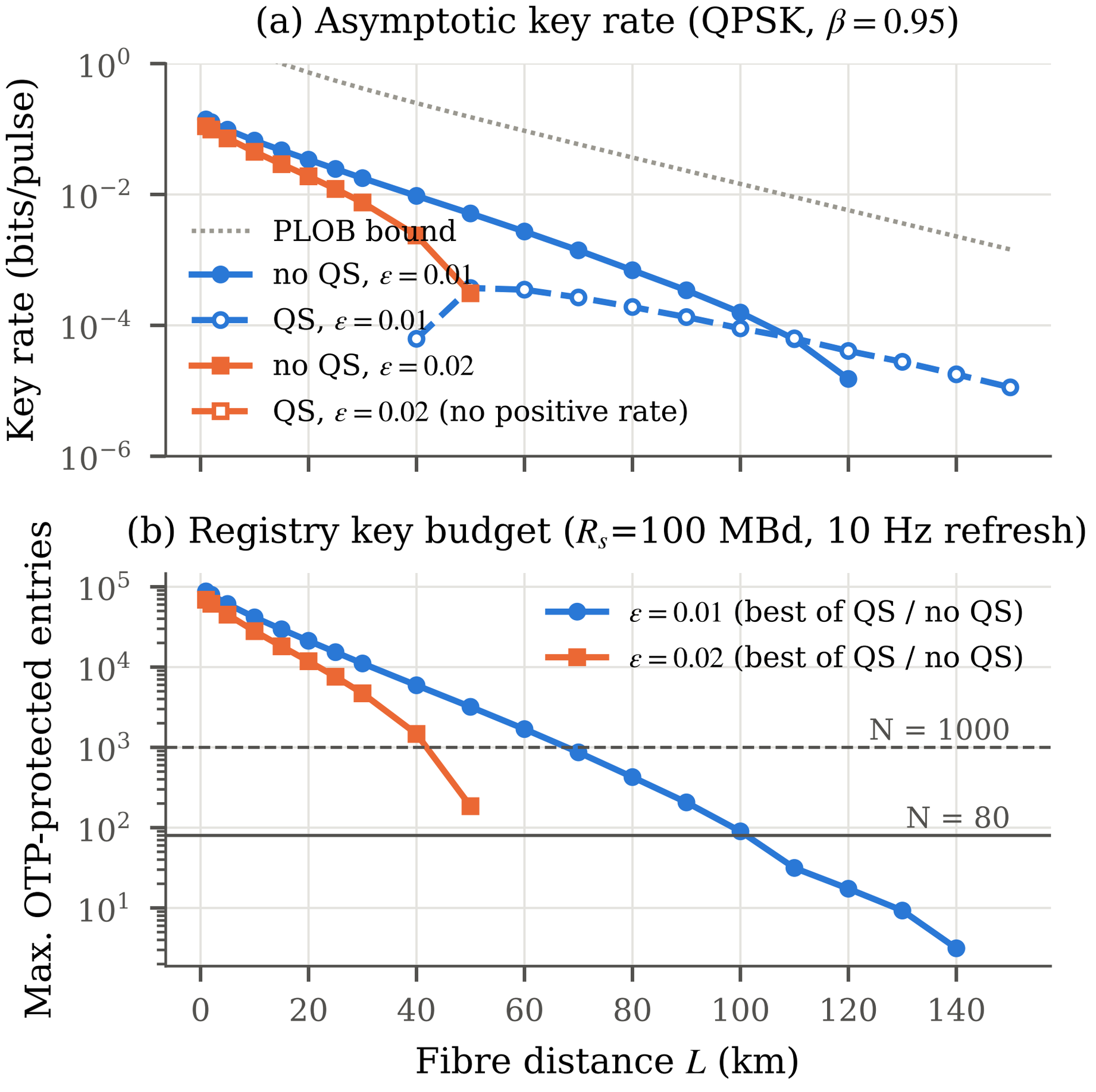
(a) Asymptotic key rate (QPSK, $\beta = 0.95$)
Key rate (bits/pulse)
10^{0}
10^{-2}
10^{-4}
10^{-6}
PLOB bound
no QS, $\varepsilon = 0.01$
QS, $\varepsilon = 0.01$
no QS, $\varepsilon = 0.02$
QS, $\varepsilon = 0.02$ (no positive rate)
(b) Registry key budget (R_s=100 MBd, 10 Hz refresh)
Max. OTP-protected entries
10^{5}
10^{4}
10^{3}
10^{2}
10^{1}
$\varepsilon = 0.01$ (best of QS / no QS)
$\varepsilon = 0.02$ (best of QS / no QS)
N = 1000
N = 80
0
20
40
60
80
100
120
140
Fibre distance L (km)